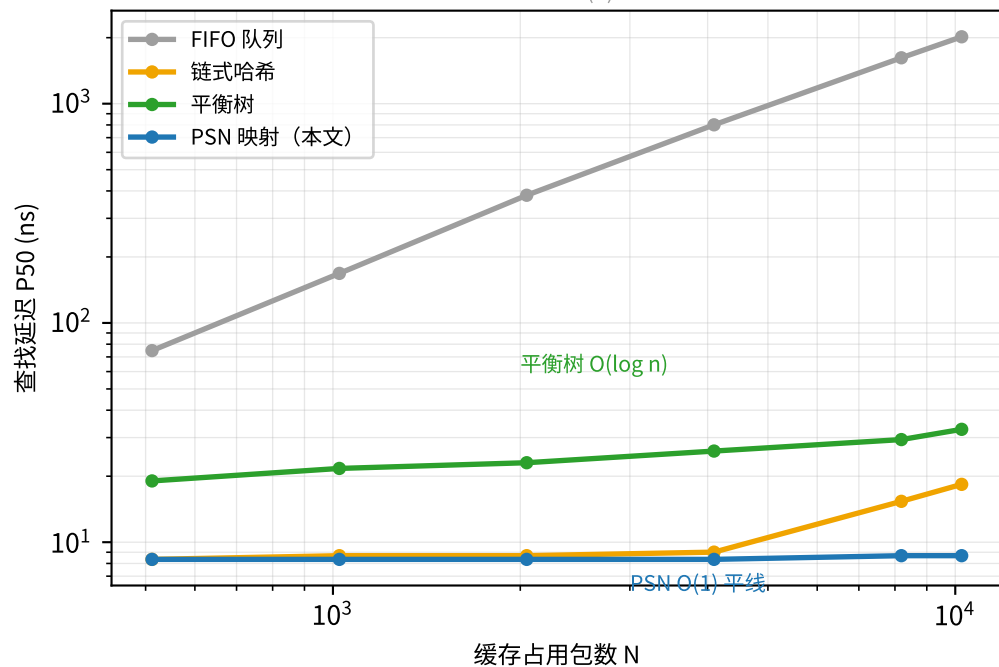
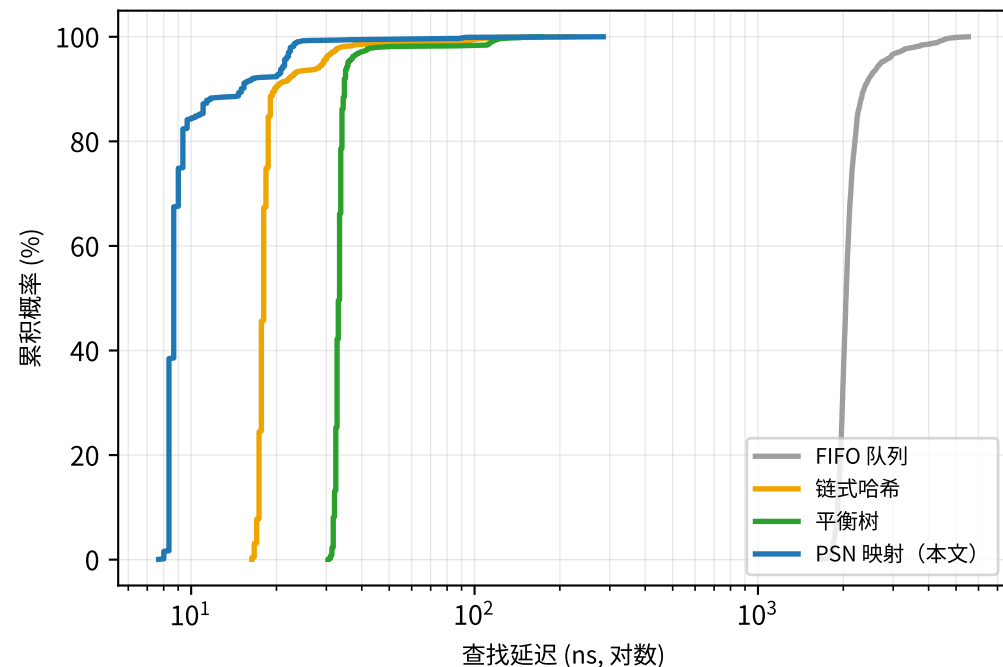


图 1 时间开销：PSN 映射 $O(1)$ 查找/存储/取回，且延迟与规模无关

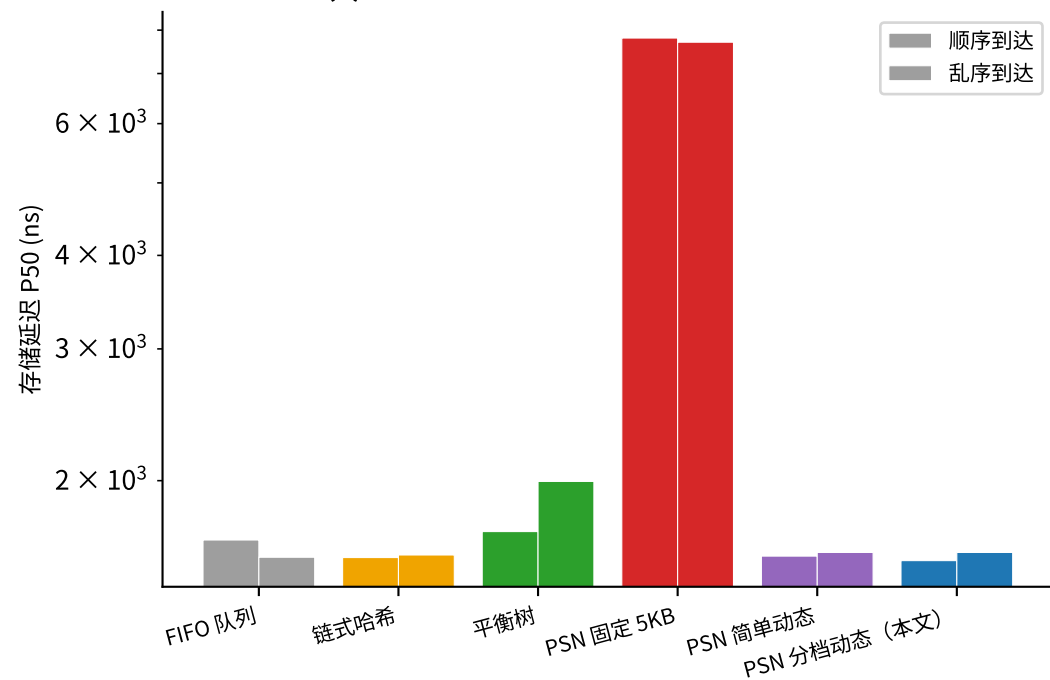
(a) 查找延迟随 N 增长 (复杂度曲线)



(b) 随机访问查找延迟 CDF



(c) 存储延迟 ($L=1024$, 固定 5KB 慢 $5\times$)



(d) 取回/丢包判定延迟 (FIFO $O(n)$ 重排 $8.8\mu s$)

